

Terms:

- 1) **Root** - The root, is the *starting* position where the first node sits.
- 2) **Parent** - To easily explain this, **Root** = starting $\rightarrow \forall n, n = \text{node that come from the root, are know as } \textit{children} \text{ and root would be the } \textit{Parent}$.
- 3) **Child** - Again, let's make it easy - above I basically explained it... but if you don't understand... the *child* in the case about would be the nodes descending from the *root*.
- 4) **Sibling** - Those who are *children* to the same *Parent* are *Siblings*.
- 5) **Descendents** - This is hard to explain without a visual but essentially if you pick a node within a tree - it's desendcants are the nodes that are *Below* it.
- 6) **Ancestor** - Ancestors are the nodes that are '*above*' the current node being looked at (which I will elaborate more on)
- 7) **Degree of node** - The degree is hard to explain without a visual - but the *Degree* is essentially the amount of
- 8) **Internal/External (Terminal/Non-Terminal)** - Nodes that leafs are *Non-Terminals* $\forall$ other nodes they are said to be *Terminals*.
- 9) **Levels** -
- 10) **Height** -
- 11) **Forest** -

$$T(n) = \frac{(2*n)_c n}{n+1}$$